

# Unrecorded eligibility changes who is eligible more than it changes the effect among them

A simulation study answering the structural part of catalog problem ELG-01

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## Abstract

**Background.** When the measurement that defines eligibility is missing for some patients, an emulation cannot tell from its records who was eligible. Catalog entry ELG-01 states that latent eligibility makes the target population unidentified without validation data. We asked whether that ambiguity changes the treatment effect materially, and how estimators with and without internal validation perform.

**Methods.** Paired full-data laws with identical observed records but different eligibility among people with unrecorded measurements (a shift of log odds  $\lambda$  of 0, log 2 or log 4) were crossed with the share unrecorded (0.20, 0.50), the expected validation sample (100, 400), a homogeneous or heterogeneous treatment effect, and whether comorbidity, the effect modifier, enters eligibility directly: 48 scenarios, 4000 replicates each, 4000 people per replicate. The registered structural contrast  $\Delta_{ID}$  compares the true risk difference among the eligible at  $\lambda = \log 4$  and  $\lambda = 0$  in the pair most likely to show a difference.

**Results.**  $\Delta_{ID}$  was -0.00066 (90% interval -0.000661 to -0.000660), inside the registered equivalence band of  $\pm 0.005$ : not materially different. The same shift moved the true prevalence of eligibility from 0.705 to 0.787, which estimators assuming eligibility is missing at random missed by up to 0.082. Every estimator's risk difference was unbiased within  $\pm 0.005$  in every scenario.

**Conclusions.** In this design, unrecorded eligibility changes how many people are eligible far more than the effect among them. Validation data are needed to count the eligible; the effect estimate is robust unless the hidden eligibility shifts the share of an effect modifier substantially.

## 1 Introduction

Eligibility criteria defined by a laboratory value or a clinical assessment are often unrecorded for part of the cohort. Restricting to recorded eligibles, or imputing eligibility under a missing-at-random assumption, then targets a population that may differ from the truly eligible one (1,2). Catalog entry ELG-01 states that without validation data the eligible population is not identified. That is true by construction: two data-generating laws can produce identical records and different eligibility among the unrecorded. Whether the difference matters for the effect estimate is a separate question, and it is the one this study asks.

## 2 Methods

The study follows the ADEMP structure (3); the protocol and decision rule were registered before any replicate was drawn.

## 2.1 Data-generating mechanism

Each replicate has 4000 people with baseline covariates age, sex, specialist access H, comorbidity C and recorded treatment intent, and a single baseline treatment decision. A clinical recording indicator R is drawn with  $P(R = 0)$  calibrated to  $m$  (0.20 or 0.50). A latent laboratory value  $G = 45 + 5T$  defines eligibility  $E = 1$  when  $G$  is at least 45, where  $T$  is logistic with linear predictor  $0.85 - 0.65Z - \kappa C + 0.25S + 0.50H + \lambda(1 - R)$ ;  $\kappa$  is 0 or 1.10. Changing  $\lambda$  changes eligibility only among people whose value is unrecorded, and leaves every observed variable bitwise identical, so paired  $\lambda$  arms are observationally equivalent. The 60-month outcome has baseline risk depending on covariates and a treatment effect of  $-0.06$  for everyone (homogeneous) or  $-0.04$  without and  $-0.12$  with comorbidity (heterogeneous). A Bernoulli validation sample with known inclusion probability reveals  $G$  for some unrecorded people.

## 2.2 Estimands and methods

The primary estimand is the 60-month risk difference among the truly eligible,  $RD_E$ ; the complete-case analysis is evaluated against the risk difference among recorded eligibles, and the difference between the two is reported as a target discrepancy. Truth was computed from ten million simulated people per law. Methods without validation: complete case; multiple imputation of  $G$  or of  $E$  under missing at random ( $M = 20$ , with an  $M = 50$  check); inverse probability of ascertainment weighting; and a pattern-mixture sensitivity analysis at assumed  $\lambda^*$  of 0, log 2 and log 4. Methods with validation: known-design two-phase weighting, validation-anchored fractional imputation and an augmented two-phase estimator. An oracle with full eligibility is the benchmark.

## 2.3 Decision rule

The structural result uses one prespecified pair: heterogeneous effect,  $m = 0.50$ ,  $\kappa = 1.10$ ,  $\lambda = \log 4$  against  $\lambda = 0$ . It is material if the 95% paired Monte Carlo interval for  $\Delta_{ID}$  lies wholly outside  $\pm 0.010$ , and not materially different if the 90% interval lies wholly within  $\pm 0.005$ , provided the observed records and the no-validation method outputs are identical across the pair. An estimator's bias is material if its 95% interval lies wholly outside  $\pm 0.010$  and equivalent if its 90% interval lies wholly within  $\pm 0.005$ .

# 3 Results

## 3.1 Structural contrast

The observed records and every no-validation method's output were identical across the pair.  $\Delta_{ID}$  was  $-0.00066$ , with Monte Carlo standard error  $4.4e-07$  and a 90% interval of  $-0.000661$  to  $-0.000660$ , wholly inside  $\pm 0.005$ . The registered classification is not materially different.

Latent eligibility did change the population. In the decisive pair, the true prevalence of eligibility was 0.705 at  $\lambda = 0$ , 0.755 at log 2 and 0.787 at log 4. The effect among the eligible hardly moved because the share of comorbid patients among them changed by less than one percentage point: with effects of  $-0.04$  and  $-0.12$ , a  $\Delta_{ID}$  of  $-0.0007$  corresponds to a shift of about 0.008 in that share.

## 3.2 Estimators

Every estimator's bias for its declared risk difference was classed equivalent in all 48 scenarios (Table 1). The largest absolute bias at  $M = 20$  or without imputation was 0.0017, and coverage ran from 0.940 to 0.974. The complete-case analysis estimates the effect among recorded eligibles, which differed from the effect among all eligibles by at most 0.0090.

Table 1: Risk-difference performance across the 48 scenarios: largest absolute bias and the range of coverage of the nominal 95% interval. The M = 50 imputation check ran on 80 replicates per scenario and is not shown.

| method                        | information   | max_abs_bias | coverage       |
|-------------------------------|---------------|--------------|----------------|
| complete_case primary         | no_validation | 0.0004       | 0.942 to 0.956 |
| mar_ipaw primary              | no_validation | 0.0009       | 0.942 to 0.954 |
| mi_continuous m20             | no_validation | 0.0011       | 0.954 to 0.973 |
| mi_indicator m20              | no_validation | 0.0011       | 0.955 to 0.974 |
| oracle_recording_ipaw primary | no_validation | 0.0009       | 0.942 to 0.955 |
| sensitivity lambda_0          | no_validation | 0.0011       | 0.944 to 0.953 |
| sensitivity lambda_log2       | no_validation | 0.0008       | 0.945 to 0.953 |
| sensitivity lambda_log4       | no_validation | 0.0007       | 0.944 to 0.953 |
| oracle primary                | oracle        | 0.0005       | 0.943 to 0.954 |
| augmented intercept_r         | validation    | 0.0008       | 0.944 to 0.962 |
| augmented primary             | validation    | 0.0007       | 0.944 to 0.957 |
| fractional omit_c_h           | validation    | 0.0011       | 0.943 to 0.953 |
| fractional primary            | validation    | 0.0005       | 0.943 to 0.953 |
| two_phase_ipw primary         | validation    | 0.0017       | 0.940 to 0.955 |

Prevalence tells the other half (Table 2). Without validation, the missing-at-random estimators and the sensitivity analysis at  $\lambda^* = 0$  recovered prevalence at  $\lambda = 0$  but underestimated it by up to 0.082 at  $\lambda = \log 4$ . The sensitivity analysis recovered it only when its assumed  $\lambda^*$  equaled the true  $\lambda$ . The three validation-based estimators were within 0.0015 at every  $\lambda$ .

Table 2: Bias in the prevalence of eligibility by the true  $\lambda$ , in the decisive setting (heterogeneous effect,  $m = 0.50$ ,  $\kappa = 1.10$ , expected validation sample 100). The complete-case row is the recorded yield, which is not a prevalence estimator.

|                               | $\lambda = 0$ | $\lambda = \log 2$ | $\lambda = \log 4$ |
|-------------------------------|---------------|--------------------|--------------------|
| augmented intercept_r         | 0.0001        | -0.0004            | -0.0004            |
| augmented primary             | 0.0002        | -0.0002            | -0.0003            |
| complete_case primary         | -0.3772       | -0.4267            | -0.4590            |
| fractional omit_c_h           | 0.0002        | -0.0003            | -0.0003            |
| fractional primary            | 0.0002        | -0.0002            | -0.0003            |
| mar_ipaw primary              | 0.0000        | -0.0495            | -0.0817            |
| mi_continuous m20             | -0.0003       | -0.0497            | -0.0820            |
| mi_indicator m20              | -0.0003       | -0.0498            | -0.0820            |
| oracle primary                | 0.0002        | 0.0000             | 0.0001             |
| oracle_recording_ipaw primary | -0.0001       | -0.0496            | -0.0818            |
| sensitivity lambda_0          | 0.0000        | -0.0495            | -0.0818            |
| sensitivity lambda_log2       | 0.0493        | -0.0001            | -0.0324            |
| sensitivity lambda_log4       | 0.0816        | 0.0322             | -0.0001            |
| two_phase_ipw primary         | -0.0009       | -0.0014            | -0.0015            |

## 4 Discussion

The structural claim of ELG-01 holds: records alone cannot tell which of two eligibility laws produced them. Whether that matters depends on the quantity. The number of eligible people is not identified, and estimators that assume eligibility is missing at random miss it by as much as the hidden shift moves it. The risk difference among the eligible differs by less than 0.001 between laws that produce the same records, because the effect differs

across patients only through comorbidity and hidden eligibility barely changes comorbidity's share. For the effect to change by 0.010 with these effect sizes, the share would have to move by about 12 percentage points.

A validation sample with known inclusion probabilities repairs both quantities. Without one, an emulation can report the effect among the eligible with some confidence and the size of the eligible population with none, unless the missing measurement is itself strongly tied to who benefits.

#### 4.1 What this study does not establish

The result is conditional on one baseline threshold criterion, a binary 60-month outcome, correctly specified models and a single effect modifier. It cannot estimate a real-world  $\lambda$  or establish empirical materiality. It does not cover external, nonprobability or unknown-probability validation, validation measured with error, time-varying eligibility, treatment switching, censoring or unmeasured treatment confounding. A hidden eligibility mechanism tied more strongly to the effect modifier than this one could change the effect materially.

#### 4.2 Review

The design was reviewed and revised before any code was written. An earlier version of the analysis called the bias-eliminated coverage function with its arguments in the wrong order; that was fixed across all studies before this run. The results have not yet been reviewed.

### Data and code availability

Code, protocol and result files: <https://github.com/choxos/TTE-open-problems/tree/main/studies/ELG-01-latent-eligibility-validation>.

### References

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